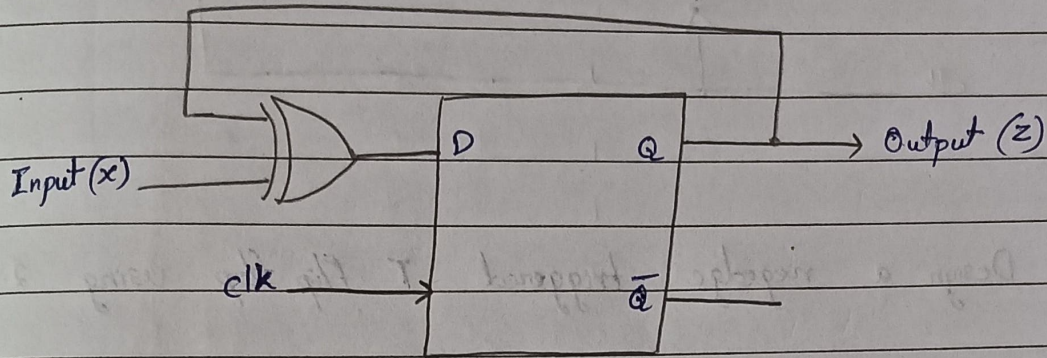


## Sequential - Assignment

- 1 Explain the behaviour of the given circuit.



from the above  $D = x \oplus Q$  and  $Q_{next} = Q$

| Q | x | $D = x \oplus Q$ | $Q_{next}$ |
|---|---|------------------|------------|
| 0 | 0 | 0                | 0          |
| 0 | 1 | 1                | 1          |
| 1 | 0 | 1                | 1          |
| 1 | 1 | 0                | 0          |

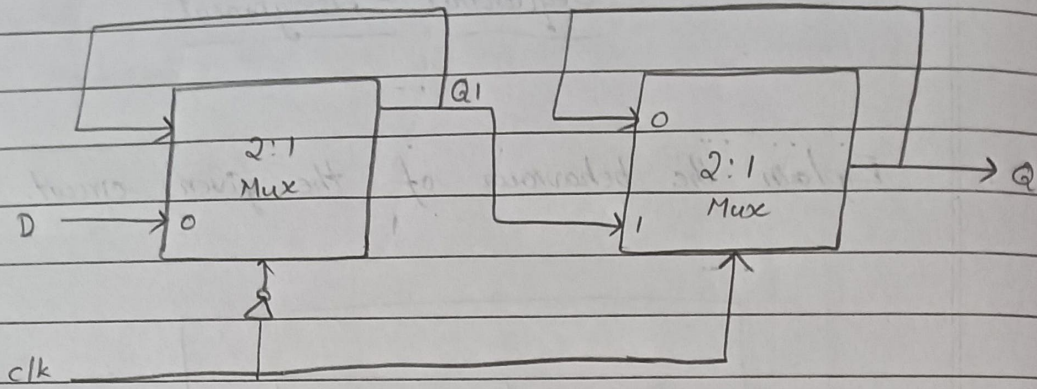
Comparing input  $x$  and the next state  $Q_{next}$  we can say that

when  $x=0$   $Q_{next} = Q$  and

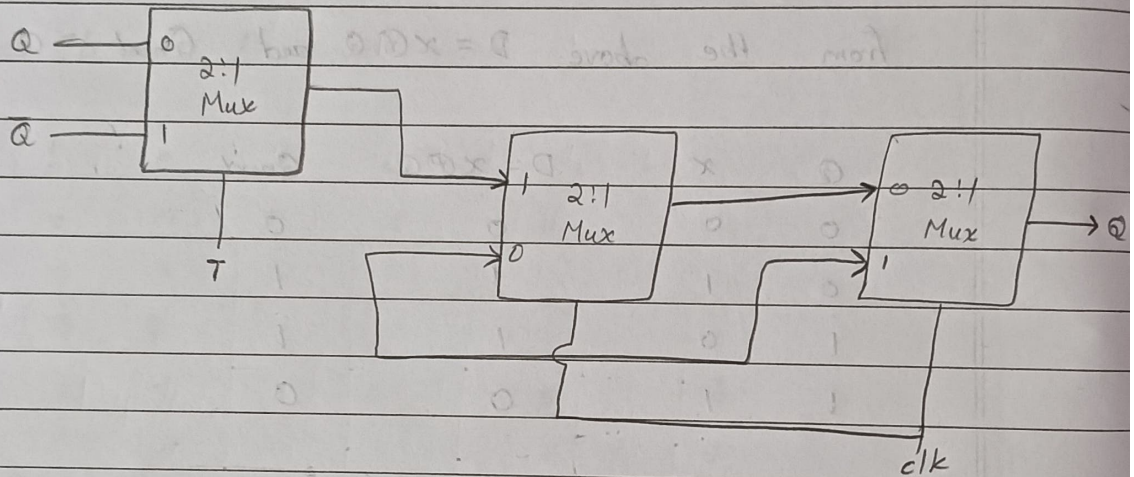
when  $x=1$   $Q_{next} = \bar{Q}$

$\therefore$  The behaviour of the given circuit is a T-flip flop.

- 2 Design a posedge triggered D-Flip flop using 2:1 Muxes only.



3 Design a negedge triggered T flip flop using 2:1 Muxes only.



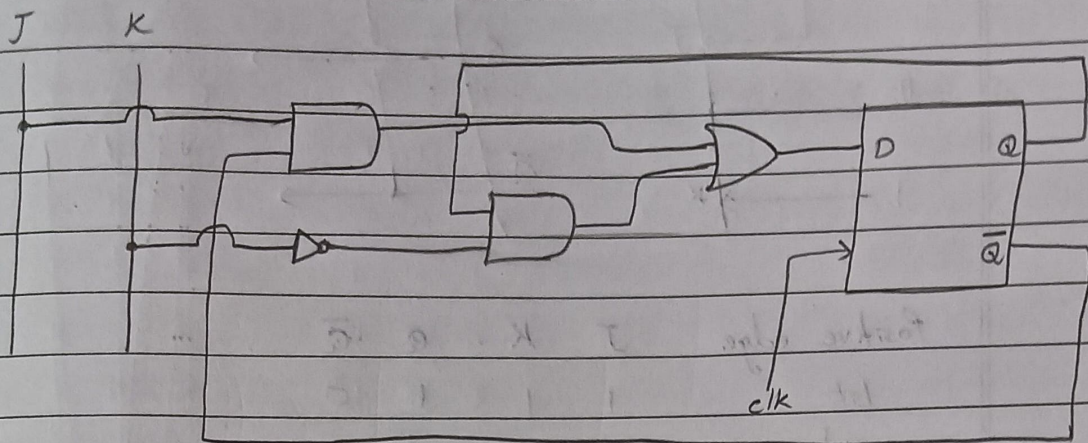
4 Convert by adding external gates.

a) A D Flip Flop to a JK Flip Flop.

|   |   | (Present state) |                   |
|---|---|-----------------|-------------------|
| J | K | Q               | Q <sub>next</sub> |
| 0 | 0 | 0               | 0                 |
| 0 | 0 | 1               | 1                 |
| 0 | 1 | 0               | 0                 |
| 0 | 1 | 1               | 0                 |
| 1 | 0 | 0               | 1                 |
| 1 | 0 | 1               | 1                 |
| 1 | 1 | 0               | 1                 |
| 1 | 1 | 1               | 0                 |

| J | KQ |    |    |    |
|---|----|----|----|----|
|   | 00 | 01 | 11 | 10 |
| 0 | 0  | 1  | 0  | 0  |
| 1 | 1  | 1  | 0  | 1  |

$$D = Q_{next} = J\bar{Q} + \bar{K}Q$$



b) A T Flip flop to a D flip flop

(Prist state)

| T | Q | Q <sub>next</sub> |
|---|---|-------------------|
| 0 | 0 | 0                 |
| 0 | 1 | 1                 |
| 1 | 0 | 1                 |
| 1 | 1 | 0                 |

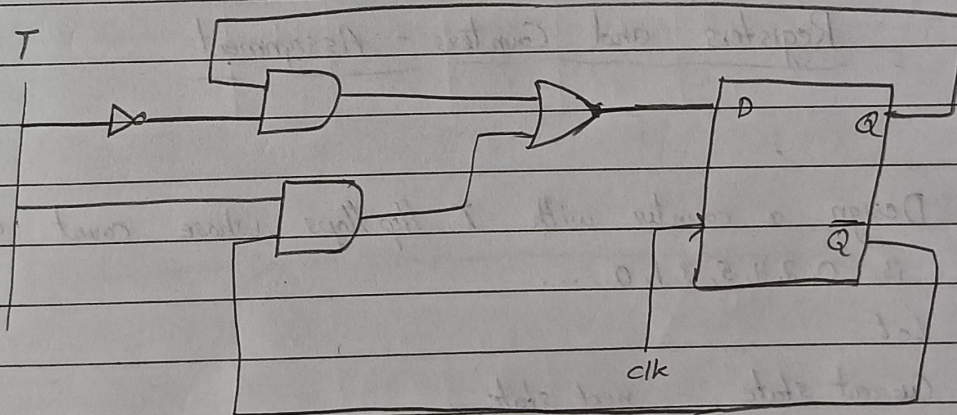
  

| T | Q |
|---|---|
| 0 | 0 |
| 0 | 1 |
| 1 | 0 |
| 1 | 1 |

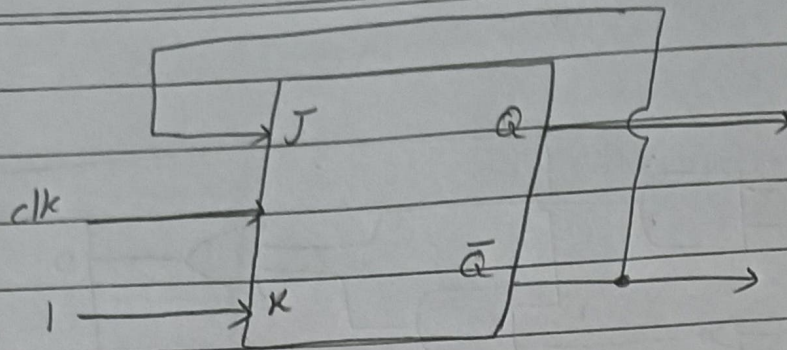
  

$$Q_{next} = \bar{T}Q + T\bar{Q}$$

or use xor gate



- 5 In a positive edge triggered JK Flip flop, we have  $J = Q$  and  $K = 1$  as shown in the figure below. Assume the flip flop was initially cleared and then clocked for 6 clock pulses, find the sequence at the Q output for those 6 clock cycles.



| Positive edge | J | K | Q | $\bar{Q}$ |
|---------------|---|---|---|-----------|
| 1st           | 1 | 1 | 1 | 0         |
| 2nd           | 0 | 1 | 0 | 1         |
| 3rd           | 1 | 1 | 1 | 0         |
| 4th           | 0 | 1 | 0 | 1         |
| 5th           | 1 | 1 | 1 | 0         |
| 6th           | 0 | 1 | 0 | 1         |

$\therefore$  The sequence Q = 1, 0, 1, 0, 1, 0